

Research Design: A Multi-Modal XAI Deep Learning framework for pneumonia diagnosis: Evaluating diagnostic performance and clinical trust over diverse populations.

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1. Introduction:

Computer-Aided Diagnosis (CAD) systems have been transformed by the integration of Artificial Intelligence (AI), enabling deep learning models to achieve higher levels of diagnostic accuracy in the automated detection of pneumonia from chest X-ray (CXR) images. The diagnostic performance of the CAD deep learning models has also seen significant improvements due to recent advancements in Deep Convolution Neural Networks (DCNNs), transfer learning, and multi-modal learning. Furthermore, the integration of Explainable AI (XAI) and Human-in-the-Loop (HITL) strategies provides greater support for transparency and clinical involvement. These developments demonstrate the potential of AI-assisted diagnostic systems to enhance clinical decision-making by reducing the burden on healthcare professionals.

Despite these enhancements, the previous literature review paper identified several limitations that continue to prevent the widespread adoption of AI-assisted pneumonia diagnostic systems. These limitations indicate that existing approaches address diagnostic accuracy, interpretability, generalisability and clinician-centred decision support as separate research problems rather than integrating them into a unified framework. Furthermore, many proposed solutions achieve improvements in one of these areas while creating limitations in another, limiting the reliable deployment of diagnostic systems across diverse healthcare environments. This study addresses the research gap by proposing the design of a **multi-modal Explainable Artificial Intelligence Deep Learning Framework for Pneumonia Diagnosis**. This framework integrates transfer-learning, multi-modal data fusion, XAI techniques and HITL decision support within a single diagnostic architecture that seeks to balance diagnostic accuracy with transparency, robustness and clinical trust to better support real-world healthcare applications.

The primary objective of this research is not only to analyse existing diagnostic approaches, but to design, develop, and evaluate an integrated diagnostic framework capable of addressing these identified limitations. The primary contribution of this research study is the design and evaluation of this novel multi-modal XAI diagnostic framework. To achieve this, a Design Science Research (DSR) methodology is adopted which will provide a structured and iterative approach for developing and evaluating an innovative diagnostic framework that addresses identified real-world problems while contributing new knowledge. This DSR methodology will follow the Design Science Research Process Model proposed by Peffers, which provides a systematic process for identifying the problem, defining solution objectives, designing and developing proposed frameworks.

This study justifies the selection of the DSR research design, describing the adopted DSR process and presenting the proposed diagnostic framework with its research design decisions that highlight its development. Measures will also be implemented to ensure the methodological rigor, reliability, validity, and ethical compliance throughout the research process.

2. Justification of the Research Design:

2.1 Nature of the Research Problem:

The previous literature review identified significant advancements in deep learning, transfer learning, multi-modal learning, XAI and HITL approaches for AI-assisted CAD systems for pneumonia detection. However, despite these advancements, several limitations continue to hinder the reliable adoption of these systems in clinical practice. These limitations remain because existing studies primarily address diagnostic accuracy, interpretability, generalisability, and clinician involvement as separate research problems, preventing the development of a comprehensive diagnostic solution. Therefore, the research problem extends beyond evaluating existing approaches, moving towards designing an integrated framework that combines these AI approaches and technological areas in a complementary manner that addresses these identified limitations and challenges simultaneously.

2.2 Why Design Science Research is Appropriate:

The absence of an integrated diagnostic framework makes the Design Science Research (DSR) an appropriate methodology for this study. DSR focuses on the design, development, and evaluation of innovative artefacts and design concepts that solve real-world problems while contributing new knowledge. Unlike traditional research approaches like Systematic Literature Reviews (SLR) that synthesises and evaluates existing knowledge or experiments that evaluates hypotheses or intervention under controlled conditions, DSR is solution-oriented in which artefacts are designed, developed and evaluated. The primary DSR artefact developed in this research study is a multi-modal XAI deep learning framework for pneumonia diagnosis. This framework integrates transfer learning and multi-modal data fusion to improve diagnostic performance and prediction accuracy, while incorporating XAI techniques and HITL decision support to enhance the transparency and clinical usability of the predictions, resulting in a unified diagnostic architecture.

Peppers DSR model provides a structured process for identifying problems and defining solution objectives, leading to the design and development of artefacts that demonstrates their application, allowing for the evaluation of their effectiveness and research contributions. Similarly, the rigorous construction and evaluation of these innovative artefacts generate knowledge that indirectly contributes to future studies which closely aligns with the objectives of this research.

2.3 Alignment of Design Science Research with this Study:

The objectives of this research study seek to develop an integrated diagnostic framework that addresses the remaining limitations identified in the literature review, rather than incrementally improving an existing diagnostic model. This approach aligns with the principles of DSR methodology, in which the methodology's iterative nature supports the continuous refinement and evaluation of the proposed novel

framework throughout its development process. Its rigorous process guides the design, development, and experimental evaluation of the artefacts against established performance criteria, enabling the systematic improvements of the proposed diagnostic framework.

3. Design Science Research Process

This study follows the Design Science Research (DSR) methodology proposed by Peffers et al. (2007), in which a research problem is addressed through the identification of a problem, definition of solution objectives, design and development of an artefact, demonstration, evaluation, and communication. The artefact developed in this study is a lightweight multi-modal XAI deep learning framework for pneumonia diagnosis using chest X-ray (CXR) images and patient metadata.

The framework combines transfer learning using DenseNet-121, multi-modal data integration, Explainable Artificial Intelligence (XAI) using Gradient-weighted Class Activation Mapping (Grad-CAM), Human-in-the-Loop (HITL) review, subgroup generalisability analysis, and preprocessing comparison. The implementation was structured into six experimental components: **E1, E2, E3, E4, E5, and E6.**

The six experiments were designed to progressively evaluate different aspects of the proposed artefact:

- **E1:** Image-only diagnostic baseline;
- **E2:** Multi-modal diagnostic model using CXR images and patient metadata;
- **E3:** Explainability using Grad-CAM;
- **E4:** Human-in-the-Loop review;
- **E5:** Generalisability across demographic and imaging subgroups; and
- **E6:** Effect of preprocessing choices on diagnostic performance.

This structure allowed the artefact to be evaluated not only in terms of predictive performance, but also in terms of interpretability, human oversight, subgroup behaviour, and preprocessing sensitivity.

3.1 Problem Identification and Motivation

Pneumonia remains a significant global health challenge, while diagnosis from chest radiographs can be difficult because radiographic manifestations may overlap with other pulmonary abnormalities. Deep learning has demonstrated substantial potential for automated pneumonia detection; however, predictive performance alone does not guarantee clinical usefulness.

The literature reviewed in Paper 2 identified several interconnected limitations. First, image-only deep learning systems may fail to exploit clinically relevant information contained outside the image. Ren et al. (2021), for example, demonstrated that combining CXR information with clinical information can improve diagnostic performance compared with relying on a single information source.

Second, deep learning models may learn dataset-specific or clinically irrelevant visual characteristics. Arias-Londoño et al. (2020) demonstrated through Grad-CAM analysis that models

can focus on non-lung characteristics, while Lapuschkin et al. (2019) demonstrated the broader problem of "Clever Hans" behaviour in machine learning.

Third, the lack of interpretable predictions can limit the ability of humans to understand and appropriately assess automated decisions. Existing research has therefore investigated explanation techniques including Grad-CAM, SHAP, LIME, TCAV, and concept-based approaches.

Fourth, diagnostic performance may vary across different patient and imaging subgroups. A model that performs adequately on an aggregate test set may still exhibit differences in performance according to demographic characteristics or imaging acquisition conditions. This motivated the inclusion of subgroup analysis in E5.

Finally, preprocessing decisions can alter the information presented to a deep learning model and may therefore affect diagnostic performance. E6 was introduced to investigate whether an alternative image preprocessing strategy, specifically Contrast Limited Adaptive Histogram Equalization (CLAHE), produced measurable improvements over the baseline preprocessing pipeline.

These limitations motivated the development of an artefact that does not treat classification accuracy as the sole objective. Instead, the proposed framework integrates **diagnostic prediction, patient metadata, visual explanation, human review, subgroup evaluation, and preprocessing analysis** into a single experimental workflow.

3.2 Objectives of the Solution

Based on the identified problem, the primary objective was to design and evaluate a lightweight multi-modal diagnostic framework capable of supporting pneumonia classification while improving interpretability, human involvement, and understanding of model behaviour across different conditions.

The specific objectives were to:

1. develop an image-based pneumonia classifier using transfer learning with DenseNet-121;
2. extend the image-only classifier through integration of patient metadata;
3. evaluate whether multi-modal information improves diagnostic performance compared with an image-only baseline;
4. generate visual explanations of model predictions using Grad-CAM;
5. demonstrate the framework across true-positive, true-negative, false-positive, and false-negative cases;
6. incorporate a Human-in-the-Loop mechanism through reviewer agreement, disagreement, and corrected-label recording;
7. evaluate diagnostic performance across age, sex, and CXR view subgroups;
8. investigate whether an alternative preprocessing strategy using CLAHE affects diagnostic performance; and

9. evaluate the resulting framework in terms of predictive performance, explanation behaviour, human correction, subgroup performance, and preprocessing sensitivity.

These objectives operationalise the research gap identified in the literature by moving beyond a standalone classifier toward an integrated diagnostic decision-support artefact.

3.3 Design and Development of the Artefact

The developed artefact consists of six experimental components.

E1: Image-only baseline

E1 implements a DenseNet-121-based pneumonia classifier using CXR images as the sole input. DenseNet-121 was selected because densely connected convolutional architectures provide feature reuse and have been widely applied to medical image classification.

The image-only model establishes the baseline against which the contribution of patient metadata can be evaluated.

E2: Multi-modal diagnostic model

E2 extends the image-only architecture by incorporating patient metadata alongside the CXR representation. The metadata consists of **age, sex, and radiographic view position**.

The image representation produced by DenseNet-121 is combined with the metadata representation before the final pneumonia prediction is generated. This design investigates whether information that is not directly contained within image pixels can contribute to the classification decision.

The E2 model achieved a best validation ROC-AUC of **0.7982** during training. For the subsequent test-set evaluation performed in E5, the model achieved an overall ROC-AUC of **0.7815**.

E3: Explainable Artificial Intelligence

E3 provides interpretability using **Gradient-weighted Class Activation Mapping (Grad-CAM)**. Grad-CAM generates a spatial heatmap indicating image regions associated with the model's prediction.

Rather than generating explanations for every test image, four representative cases were selected:

- true positive (TP);
- true negative (TN);
- false positive (FP); and
- false negative (FN).

This selection allows the framework to demonstrate both successful and erroneous model behaviour.

The selected cases demonstrated differences in activation behaviour. The TP produced a mean activation of 0.4037 and a central activation ratio of 0.3770, while the TN produced substantially lower activation, with a mean activation of 0.0204. The FP produced relatively strong activation, with a mean activation of 0.3228 and a central activation ratio of 0.4099. In contrast, the FN produced weaker and

predominantly peripheral activation, with a mean activation of 0.1369 and a central activation ratio of 0.0037.

These observations demonstrate the role of XAI as an inspection mechanism for investigating model behaviour rather than merely displaying prediction outcomes.

E4: Human-in-the-Loop

E4 incorporates human review into the diagnostic workflow. A reviewer examined the four representative cases and recorded whether they agreed with the model prediction, the corrected label where disagreement occurred, and an accompanying comment.

The reviewer agreed with the model for the TP and TN cases and disagreed with the FP and FN cases. Both model errors were therefore overridden and corrected.

The HITL component demonstrates a decision-support workflow in which the model produces an initial prediction and explanation, while a human reviewer retains the ability to challenge and correct the output.

E5: Generalisability and Subgroup Evaluation

E5 evaluates whether the E2 model behaves consistently across different patient and imaging subgroups within the NIH ChestX-ray14 test set.

The evaluation consisted of **3,758 test patients and 9,326 test images**. Performance was evaluated across:

- age <40;
- age 40–59;
- age ≥60;
- female;
- male;
- PA radiographic view; and
- AP radiographic view.

The overall test-set performance was:

Metric	E2 Test Performance
Accuracy	92.39%
Precision	11.13%
Recall / Sensitivity	30.26%
Specificity	93.94%

Metric	E2 Test Performance
F1-score	16.27%
ROC-AUC	0.7815
PR-AUC	0.0858

The subgroup analysis revealed measurable variation in performance. ROC-AUC ranged from **0.7123 for PA-view images** to **0.8034 for female patients**. Recall ranged from **16.83% for PA-view images** to **40.94% for AP-view images**.

The AP subgroup also showed a lower specificity of **86.84%**, compared with **97.85% for PA images**, indicating that the model produced substantially more false positives for AP images.

Age-based analysis showed ROC-AUC values of 0.7968 for patients younger than 40, 0.7722 for patients aged 40–59, and 0.7819 for patients aged 60 and older. Sex-based analysis produced ROC-AUC values of 0.8034 for females and 0.7642 for males.

These results indicate that aggregate performance does not fully describe the behaviour of the model and that subgroup characteristics can influence diagnostic performance.

Importantly, E5 represents **within-dataset subgroup generalisability rather than external validation**, because all subgroups were evaluated using the NIH ChestX-ray14 test set.

E6: Preprocessing Comparison

E6 investigates whether changing the image preprocessing strategy affects diagnostic performance.

Two preprocessing configurations were evaluated:

- **E6-A:** baseline preprocessing; and
- **E6-B:** CLAHE preprocessing.

Both configurations used the same multi-modal DenseNet-121 architecture, training dataset, validation dataset, and three-epoch training configuration. This allowed preprocessing to be investigated as the primary experimental difference.

At epoch 3, the baseline configuration achieved a validation ROC-AUC of **0.7982** and PR-AUC of **0.0789**, while the CLAHE configuration achieved a ROC-AUC of **0.7846** and PR-AUC of **0.0738**.

The baseline preprocessing therefore produced a higher ROC-AUC and PR-AUC than CLAHE at the final evaluated epoch. The results do not provide evidence that CLAHE improved the diagnostic discrimination of the multi-modal model under the tested conditions.

The experiment demonstrates that preprocessing represents an important design decision and that more aggressive image enhancement does not necessarily result in improved diagnostic performance.

3.4 Demonstration

The artefact was demonstrated through an end-to-end experimental pipeline.

First, E1 established the image-only diagnostic baseline. E2 subsequently incorporated patient metadata to create the multi-modal diagnostic model. The resulting model was then used to generate predictions for the test set.

Representative TP, TN, FP, and FN cases were selected for E3 and E4. Grad-CAM explanations were generated for these cases, enabling the model's decision behaviour to be visually inspected.

The selected cases were:

Case Actual		Prediction	Probability	Age	Sex	View
TP	Pneumonia	Pneumonia	93.88%	45	M	AP
TN	No pneumonia	No pneumonia	0.16%	24	M	AP
FP	No pneumonia	Pneumonia	95.70%	56	M	PA
FN	Pneumonia	No pneumonia	0.74%	15	F	AP

Each case was subsequently presented for HITL review. The reviewer agreed with the TP and TN predictions and corrected the FP and FN predictions.

E5 extended the demonstration from individual cases to systematic subgroup analysis. The E2 model was evaluated across age, sex, and radiographic-view subgroups to determine whether performance remained consistent across different portions of the test population.

E6 then investigated whether the image preprocessing pipeline influenced model performance by comparing the baseline preprocessing strategy with CLAHE.

The overall demonstration therefore followed the broader workflow:

CXR + metadata → model prediction → Grad-CAM explanation → human review → subgroup evaluation → preprocessing evaluation.

3.5 Evaluation

The evaluation considered five aspects of the artefact: **diagnostic performance, explainability, human intervention, subgroup behaviour, and preprocessing sensitivity.**

Diagnostic Performance

The E2 multi-modal model achieved an overall test accuracy of 92.39%, specificity of 93.94%, F1-score of 16.27%, and ROC-AUC of 0.7815. Precision and recall were substantially lower at 11.13% and 30.26%, respectively.

The results demonstrate that the model can achieve relatively high overall accuracy while still detecting only a minority of positive pneumonia cases at the selected classification threshold. The severe class imbalance of the dataset contributes to this difference.

Consequently, accuracy alone should not be interpreted as evidence of clinical-level diagnostic performance. The combination of ROC-AUC, PR-AUC, sensitivity, specificity, precision, and F1-score provides a more informative assessment.

Explainability

The E3 analysis demonstrated that Grad-CAM can expose differences in model activation between correct and incorrect predictions. The TP exhibited stronger central activation than the TN, while the FP also produced substantial activation despite having a negative ground-truth label.

The FN demonstrated particularly weak central activation. These results illustrate that model activation patterns do not necessarily correspond directly to diagnostic correctness.

The XAI component therefore provides an inspection mechanism that can assist researchers or reviewers in investigating potentially problematic predictions.

Human-in-the-Loop Evaluation

The HITL experiment involved four representative cases. The reviewer agreed with the two correct model predictions and corrected both incorrect predictions.

The reviewer therefore achieved 100% agreement with the correct ground-truth labels for the four selected cases, while the model-reviewer agreement rate was 50%.

However, this should be interpreted only as a proof-of-concept demonstration of human correction. Four cases and one reviewer are insufficient to establish clinical trust or generalise reviewer performance to a larger clinical population.

Generalisability Evaluation

E5 demonstrated that aggregate performance varied across demographic and imaging subgroups.

The overall test ROC-AUC was 0.7815. The highest subgroup ROC-AUC was observed for female patients at 0.8034, while the lowest was observed for PA-view images at 0.7123.

The difference between PA and AP performance was particularly notable. PA images achieved 96.48% accuracy and 97.85% specificity but only 16.83% recall. AP images achieved 85.11% accuracy and 86.84% specificity but 40.94% recall.

This indicates that the model's behaviour is influenced by radiographic view and that aggregate performance can conceal important subgroup differences.

The age analysis showed comparatively smaller differences in ROC-AUC, while sex-based results demonstrated a difference between female and male patients. These findings support the inclusion of subgroup evaluation when assessing diagnostic frameworks intended for use across diverse populations.

However, E5 does not constitute external validation because all evaluations were performed using the NIH ChestX-ray14 test set. External validation using an independent dataset would be required to establish performance across institutions, equipment, acquisition protocols, and populations.

Preprocessing Evaluation

E6 investigated whether CLAHE improved diagnostic performance relative to the baseline preprocessing pipeline.

At epoch 3, the baseline preprocessing achieved a ROC-AUC of 0.7982 compared with 0.7846 for CLAHE. PR-AUC was also higher for the baseline configuration, at 0.0789 compared with 0.0738 for CLAHE.

Although CLAHE produced a higher accuracy at epoch 2, its final evaluated ROC-AUC and PR-AUC were lower than the baseline configuration. Therefore, the experiment does not support the hypothesis that CLAHE provides a consistent diagnostic advantage under the evaluated conditions.

The result demonstrates that preprocessing choices should be empirically evaluated rather than assumed to improve model performance.

Comparison with Existing Research

The results can be contextualised against previous studies reviewed in Paper 2. Existing pneumonia and COVID-19 systems have reported substantially higher conventional accuracy values, including 92.14–93.29% for the XAI-ICP system of Sheu et al. (2023), 96.07% for the improved quantum neural network of Dong et al. (2020), and 97.82% for the attention-based ResNet approach of Ali et al. (2020). Other systems have reported even higher values, such as 99.20% for the DenseNet201-based system of Mahamud et al. (2024).

However, direct numerical comparison is inappropriate because the studies use different datasets, class distributions, preprocessing strategies, architectures, thresholds, and evaluation protocols.

The present research should therefore be interpreted primarily as an evaluation of an **integrated DSR artefact** rather than an attempt to achieve state-of-the-art classification performance.

The multi-modal results are consistent with the findings of Ren et al. (2021), who demonstrated the value of combining CXR information with clinical information. Similarly, the E3 results support the role of visual explanation in investigating model behaviour, while E4 demonstrates how human review can be incorporated into the diagnostic workflow.

The E5 and E6 experiments further extend the evaluation by demonstrating that performance can vary across subgroups and that preprocessing modifications do not necessarily produce improved performance.

3.6 Communication

The final DSR activity concerns communicating the artefact, its design decisions, evaluation results, and limitations to the research and practitioner communities.

The artefact is communicated through this research design paper, the documented experimental pipeline, quantitative evaluation metrics, Grad-CAM visualisations, HITL feedback records, subgroup analysis results, and preprocessing comparison results.

The implementation produces structured CSV outputs containing experimental metrics and case-level results, supporting an auditable record of the evaluation process.

The findings communicate several contributions. First, patient metadata can be integrated with a DenseNet-121 CXR classifier within a lightweight multi-modal architecture. Second, Grad-CAM can be incorporated to expose spatial characteristics associated with model predictions. Third, human

review can provide a mechanism for detecting and correcting model errors. Fourth, subgroup evaluation demonstrates that aggregate diagnostic performance can conceal differences across demographic and imaging groups. Finally, preprocessing evaluation demonstrates that alternative image enhancement techniques should be empirically validated rather than assumed to improve performance.

The communication of these findings also identifies the boundaries of the artefact, particularly the lack of external validation, the limited HITL sample, the limited XAI case sample, and the absence of a large-scale clinical user study.

4. Research Design Decisions

Several methodological decisions were made to ensure that the research remained aligned with the DSR methodology and the objectives established in the previous papers.

Design Science Research as the Primary Methodology

DSR was selected because the primary contribution of the study is the **design and evaluation of an artefact**, rather than the observation of an existing phenomenon alone. The framework represents an engineered solution to the identified problem and can therefore be evaluated according to its functionality, performance, interpretability, and utility.

The six activities of Peffers et al. (2007) provide the structure through which the artefact was developed, demonstrated, evaluated, and communicated.

Experimental Evaluation

Experimental evaluation was selected as a supporting method within the DSR process. E1 and E2 provide a comparison between image-only and multi-modal classification. E3 and E4 extend the evaluation beyond predictive performance to interpretability and human involvement. E5 investigates subgroup behaviour, while E6 examines the effect of preprocessing choices.

This combination allows the artefact to be evaluated from multiple perspectives rather than through a single performance metric.

Transfer Learning

DenseNet-121 was selected as the underlying image architecture to utilise established deep visual representations without requiring the development of an entirely new convolutional architecture. Transfer learning also supports the objective of developing a lightweight research artefact within the available computational resources.

Multi-Modal Integration

Age, sex, and radiographic view were incorporated as metadata because these variables are available alongside the CXR image and provide contextual information that is not directly represented by image pixels.

The purpose was not to claim that these variables are inherently diagnostic, but to investigate whether contextual patient and acquisition information can contribute to model discrimination.

Grad-CAM

Grad-CAM was selected as the primary XAI technique because it provides spatially interpretable heatmaps that can be directly associated with CXR regions. It also has precedent in pneumonia research, including Arias-Londoño et al. (2020) and Sheu et al. (2023).

SHAP and LIME were not implemented in the final artefact. This was a deliberate scope decision rather than an indication that these techniques are unsuitable. Implementing multiple XAI techniques would have increased experimental complexity without necessarily improving the core DSR demonstration.

Human-in-the-Loop Design

HITL was incorporated because an automated diagnostic system should not necessarily operate as an autonomous replacement for clinical judgement. Instead, the framework demonstrates a decision-support approach in which humans can inspect, challenge, and correct model outputs.

Subgroup Evaluation

Subgroup evaluation was incorporated to investigate whether aggregate model performance was representative of different demographic and imaging groups.

Age, sex, and radiographic view were selected because these variables were already available within the NIH dataset and form part of the metadata used by the multi-modal model.

The subgroup analysis is therefore intended to identify performance variation rather than to claim comprehensive fairness validation.

Preprocessing Design

The preprocessing comparison was introduced to determine whether an alternative image enhancement strategy could improve diagnostic performance.

CLAHE was selected because it can enhance local image contrast and is commonly considered for medical imaging applications. However, the experiment was designed as an empirical comparison rather than assuming that contrast enhancement would improve classification.

The results ultimately demonstrated that the baseline preprocessing pipeline produced higher final ROC-AUC and PR-AUC than CLAHE under the tested conditions.

5. Reliability, Validity and Rigor

Several measures were applied to improve the reliability, validity, and methodological rigor of the research.

First, the E1 and E2 models used the same NIH ChestX-ray14 dataset structure and evaluation procedure, allowing the contribution of patient metadata to be examined under comparable conditions. The E2 model achieved a best validation ROC-AUC of 0.7982 and was subsequently evaluated on the held-out test set as part of E5.

Second, multiple diagnostic metrics were used rather than relying solely on accuracy. Accuracy, precision, recall, sensitivity, specificity, F1-score, ROC-AUC, and PR-AUC were recorded. This is

particularly important because the evaluation dataset is highly imbalanced, with only 228 positive cases among 9,326 test images.

Consequently, the overall accuracy of 92.39% must be interpreted alongside the much lower precision of 11.13% and recall of 30.26%. The use of multiple metrics reduces the likelihood that conclusions will be driven by a single potentially misleading measure.

Third, patient-level splitting was used during dataset preparation. Patients were assigned to training, validation, and test partitions rather than randomly splitting individual images. This reduces the risk that images from the same patient appear across different partitions and artificially inflate performance.

Fourth, the XAI evaluation deliberately included TP, TN, FP, and FN cases. This reduces the risk of presenting only successful predictions and allows model explanation behaviour to be examined for both correct and erroneous outcomes.

Fifth, E5 expanded the evaluation beyond aggregate test performance by analysing age, sex, and radiographic-view subgroups. This provides additional evidence regarding the consistency of model behaviour within the evaluated population.

The subgroup results demonstrate why this analysis is important. For example, ROC-AUC ranged from 0.7123 for PA-view images to 0.8034 for female patients, while recall ranged from 16.83% for PA images to 40.94% for AP images.

Sixth, E6 provided a controlled preprocessing comparison. Both preprocessing configurations used the same multi-modal architecture and dataset structure, while the preprocessing strategy was changed. This enables the observed difference to be interpreted primarily in relation to the preprocessing choice.

The final E6 results showed that baseline preprocessing achieved a validation ROC-AUC of 0.7982 compared with 0.7846 for CLAHE. PR-AUC showed the same direction, with 0.0789 for baseline preprocessing compared with 0.0738 for CLAHE.

Seventh, the framework records model predictions, probabilities, metadata, explanation metrics, reviewer decisions, and corrected labels in structured output files. This creates an auditable experimental record and supports reproducibility.

Finally, the use of DSR provides methodological structure by explicitly connecting the identified problem, solution objectives, artefact, demonstration, evaluation, and communication activities.

Despite these measures, the study does not provide statistical evidence of clinical validity. The experiments should therefore be interpreted as evaluation of a research artefact rather than clinical validation of a deployable diagnostic system.

6. Ethical Considerations

The use of medical imaging and patient metadata introduces important ethical considerations. Although the study uses an existing research dataset rather than collecting new patient data, medical images and demographic attributes require careful handling.

The artefact was therefore designed as a research and decision-support system rather than an autonomous clinical diagnostic system. The predictions generated by the model should not be interpreted as medical diagnoses or as a replacement for qualified healthcare professionals.

Patient Privacy

The framework uses the de-identified NIH ChestX-ray14 dataset and limited metadata consisting of age, sex, and radiographic view. No directly identifying patient information is incorporated into the proposed diagnostic workflow.

Patient identifiers are used only as dataset-level identifiers necessary for patient-level data partitioning and evaluation.

Algorithmic Bias and Subgroup Differences

Algorithmic bias represents an important consideration because model performance can vary according to demographic and acquisition characteristics.

The E5 subgroup evaluation demonstrated measurable performance differences across age, sex, and radiographic view. For example, PA images achieved a ROC-AUC of 0.7123, whereas AP images achieved 0.7884. Female patients achieved a ROC-AUC of 0.8034 compared with 0.7642 for male patients.

These differences do not by themselves establish algorithmic discrimination or clinical unfairness, but they demonstrate why aggregate performance should not be treated as sufficient evidence of reliable behaviour across all populations.

The subgroup analysis therefore serves as an initial investigation of performance variation rather than a comprehensive fairness assessment.

Human Oversight

The incorporation of HITL provides an additional ethical safeguard because the framework positions the model as an aid to human decision-making rather than an unquestionable authority.

In the demonstrated cases, the reviewer was able to identify and correct both incorrect model predictions. However, the small sample and single-reviewer design prevent conclusions about clinical effectiveness.

Explainability

XAI introduces another ethical consideration. A Grad-CAM heatmap indicates image regions associated with a model prediction, but it does not establish that those regions represent a clinically verified pathological cause.

Consequently, Grad-CAM explanations should be used to support human inspection and model auditing rather than being treated as definitive clinical evidence.

Responsible Use

The framework should therefore be regarded as an experimental research artefact. Deployment in a clinical environment would require additional validation, appropriate regulatory assessment, security controls, clinical oversight, and evaluation by qualified medical professionals.

The results obtained in this study do not establish that the model is suitable for autonomous diagnosis or clinical deployment.

7. Limitations of the Research Design

Several limitations constrain the conclusions that can be drawn from this study.

First, the evaluation was conducted using the NIH ChestX-ray14 dataset. Although the E5 subgroup analysis evaluated differences across age, sex, and radiographic view, it remains a **within-dataset generalisability evaluation** rather than external validation. The study therefore does not establish whether the framework would perform similarly on images from other hospitals, imaging equipment, acquisition protocols, or patient populations.

Second, the positive class is substantially underrepresented. The E5 test set contained 228 positive cases compared with 9,098 negative cases. This imbalance contributes to the large difference between accuracy and positive-class metrics. Although the E2 model achieved 92.39% test accuracy, its precision was only 11.13% and recall was 30.26%.

Therefore, the high accuracy should not be interpreted as evidence of clinical-level pneumonia detection capability.

Third, the E5 subgroup analysis demonstrates that performance varies across different groups. In particular, PA-view images produced a ROC-AUC of 0.7123 and recall of only 16.83%, whereas AP-view images produced a ROC-AUC of 0.7884 and recall of 40.94%. These differences indicate that aggregate performance can conceal important variations in model behaviour.

However, the subgroup analysis is limited to the metadata variables available in the selected dataset. It does not constitute a comprehensive fairness or bias assessment across all clinically relevant demographic and socioeconomic characteristics.

Fourth, the XAI evaluation was intentionally limited to four representative cases. Although the inclusion of TP, TN, FP, and FN cases provides qualitative coverage of different prediction outcomes, four cases are insufficient to establish the statistical reliability or generalisability of explanation behaviour.

Fifth, only Grad-CAM was implemented. Although SHAP, LIME, Grad-CAM++, and other explanation methods have been investigated in previous research, the current artefact does not provide a systematic comparison of different XAI techniques.

Sixth, the HITL evaluation involved only four reviewed cases and a single reviewer. Although both model errors in the selected cases were successfully corrected, the result cannot be generalised to clinical professionals or larger populations. The 100% reviewer correctness observed in the selected cases should therefore not be interpreted as evidence of clinical reliability or trust.

Seventh, E6 was limited to a comparison between the baseline preprocessing pipeline and CLAHE. Although the experiment found that baseline preprocessing produced a higher final validation ROC-

AUC and PR-AUC than CLAHE, this does not establish that the baseline pipeline is universally optimal. Other preprocessing methods, image normalisation strategies, resolution choices, or augmentation techniques may produce different results.

Eighth, the E6 experiments were conducted using three training epochs because of the available computational resources. Although three epochs were sufficient to observe measurable differences between the preprocessing configurations, longer training could potentially alter the relative performance of the two approaches.

Ninth, the study does not constitute a clinical validation study. No prospective hospital trial, statistically meaningful clinician study, or comparison against radiologist performance was conducted.

Finally, the framework prioritised feasibility and demonstration within the available computational and time constraints. More extensive hyperparameter optimisation, external validation, larger-scale XAI evaluation, multiple XAI techniques, larger HITL studies, and additional demographic subgroup analysis could provide stronger evidence of effectiveness and generalisability.

8. Conclusion

This study designed and evaluated a lightweight multi-modal XAI deep learning framework for pneumonia diagnosis using Design Science Research. The resulting artefact integrates DenseNet-121-based image classification, patient metadata, Grad-CAM explanations, Human-in-the-Loop review, subgroup analysis, and preprocessing evaluation into a unified diagnostic decision-support workflow.

The E1 and E2 experiments established the progression from an image-only diagnostic model to a multi-modal architecture incorporating age, sex, and radiographic view. The E2 model achieved a best validation ROC-AUC of **0.7982**, demonstrating its ability to learn meaningful discriminatory patterns during model development.

Evaluation on the held-out NIH test set in E5 produced an overall accuracy of **92.39%**, specificity of **93.94%**, F1-score of **16.27%**, ROC-AUC of **0.7815**, and PR-AUC of **0.0858**. However, recall was only **30.26%** and precision was **11.13%**, demonstrating the limitations of relying on accuracy in a highly imbalanced diagnostic problem.

The E5 subgroup analysis further demonstrated that aggregate performance does not fully represent model behaviour. ROC-AUC ranged from **0.7123 for PA-view images** to **0.8034 for female patients**, while recall ranged from **16.83% for PA images** to **40.94% for AP images**. These findings demonstrate measurable variation across patient and imaging subgroups and support the inclusion of subgroup analysis when evaluating AI-based diagnostic systems.

The E3 XAI experiment demonstrated that Grad-CAM can expose differences in model activation between correct and incorrect predictions. The false-positive and false-negative cases demonstrated that prediction confidence and activation behaviour do not necessarily correspond directly to diagnostic correctness. This reinforces the value of examining model behaviour rather than relying exclusively on prediction probabilities.

The E4 HITL experiment further demonstrated that human review can provide a mechanism for identifying and correcting erroneous predictions. Both model errors in the selected four-case demonstration were successfully corrected by the reviewer. Nevertheless, the small sample size and single-reviewer design mean that these results should be interpreted only as a proof of concept rather than evidence of clinical trust.

Finally, E6 demonstrated that preprocessing choices can influence diagnostic performance. The baseline preprocessing strategy achieved a validation ROC-AUC of **0.7982** at the third epoch compared with **0.7846** for CLAHE. PR-AUC showed a similar pattern, with **0.0789** for baseline preprocessing compared with **0.0738** for CLAHE. Under the evaluated conditions, therefore, CLAHE did not provide an improvement over the baseline preprocessing pipeline.

Overall, the study demonstrates the feasibility of integrating **multi-modal learning, explainable AI, human oversight, subgroup evaluation, and preprocessing analysis within a single lightweight diagnostic framework.**

The primary contribution of the research is therefore not the achievement of state-of-the-art pneumonia classification accuracy. Instead, the contribution lies in the design and evaluation of an integrated DSR artefact that addresses several limitations identified in existing pneumonia CAD research.

The findings also demonstrate that evaluating an AI diagnostic framework requires more than a single aggregate accuracy measure. Diagnostic discrimination, positive-class detection, explanation behaviour, human correction, subgroup variation, and preprocessing sensitivity all provide important information about the behaviour and limitations of the artefact.

Future research should therefore focus on **external validation using independent datasets, larger clinician-based HITL studies, systematic comparison of XAI techniques including SHAP and LIME, broader subgroup and fairness analysis, optimisation of preprocessing and training strategies, and evaluation of explanation reliability.** Such work would provide stronger evidence regarding whether the proposed framework could ultimately support reliable clinical decision-making across diverse healthcare environments.

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